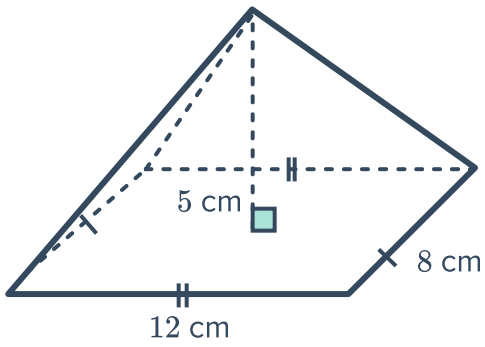
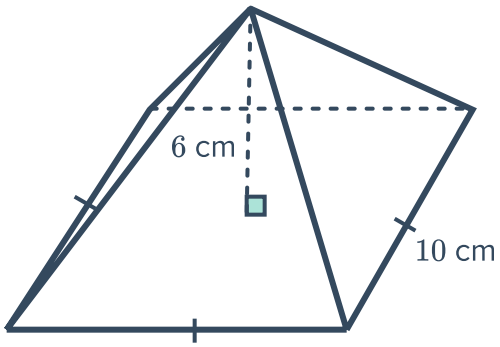
Volume of pyramids and cones

1 Find the volume of the following pyramids, correct to two decimal places if necessary:

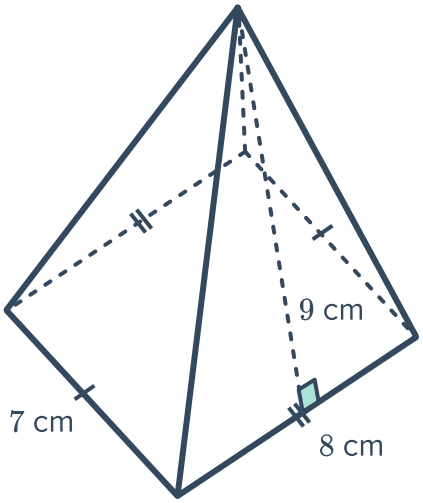
a



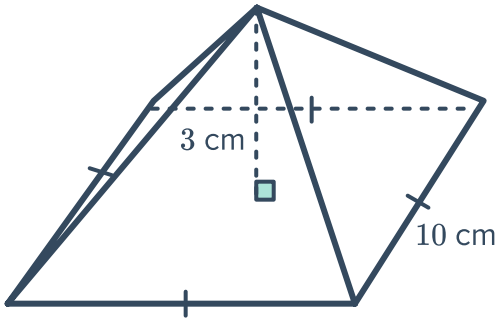
b



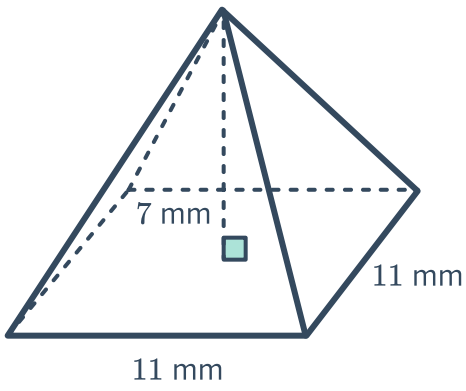
c



d

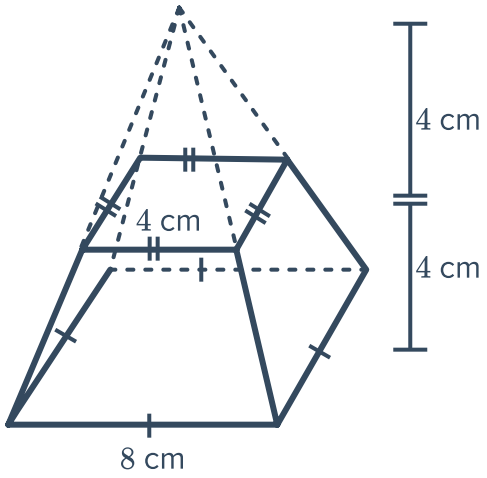


e



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2 A small square pyramid of height $4\ \text{cm}$ was removed from the top of a large square pyramid of height $8\ \text{cm}$ forming the solid shown. Find the exact volume of the solid.



3 Suppose one particular pyramid in Egypt has a square base of side length $231\ \text{m}$ and with a perpendicular height of $144\ \text{m}$. Find the volume of the pyramid, correct to the nearest m^3.

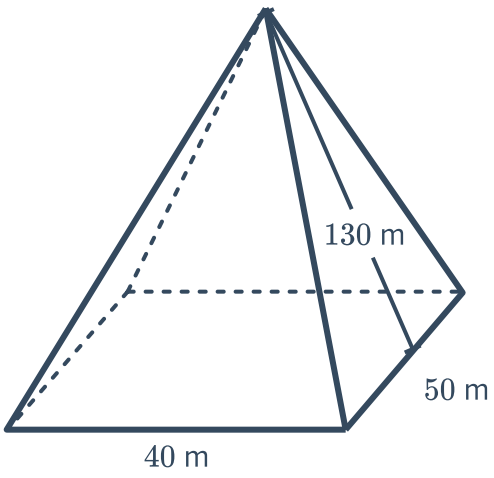
4 A rectangular pyramid has a volume of $288\ \text{cm}^3$. The base has a width of $12\ \text{cm}$ and length $6\ \text{cm}$. Find the height of the pyramid.

5 Consider the following right pyramids:

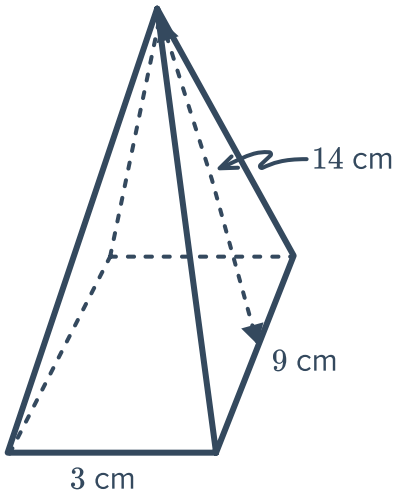
a Find the perpendicular height of the pyramid, correct to two decimal places.

b Hence, find the volume of the pyramid, correct to two decimal places.

a



b



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6 Some very famous right pyramids are the Egyptian Pyramids.
The diagram shows the Great Pyramid. It has a square base of side length $230\ \text{m}$ and a vertical height of $146\ \text{m}$. Find the volume of the Great Pyramid, correct to the nearest tenth.



7 Find the volume of the following right pyramids:

a A right pyramid with a square base of $24\ \text{mm}$ and a perpendicular height of $24\ \text{mm}$.

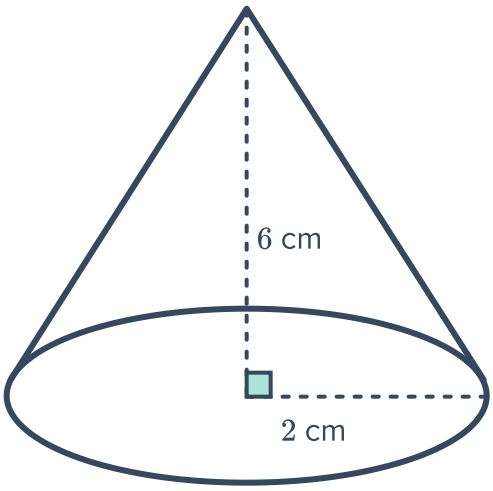
b A right pyramid with a square base of $21\ \text{mm}$ and a vertical height of $27\ \text{mm}$.

8 A right square pyramid has a base of length $12\ \text{mm}$ and a volume $1248\ \text{mm}^3$. Find the height of the pyramid.

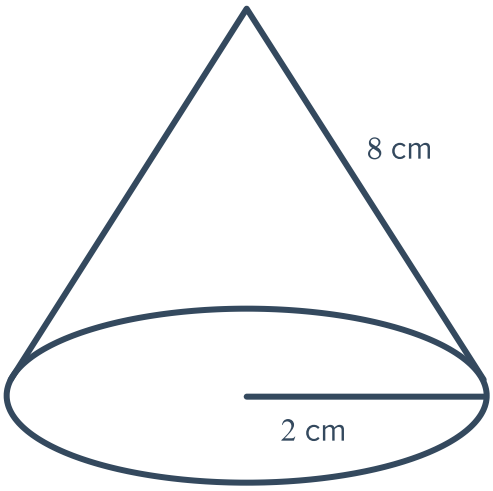
9 A right square pyramid has a height of $24\ \text{cm}$ and a volume $2592\ \text{cm}^3$. Find the base length of the pyramid.

10 Find the volume of the following cones, correct to two decimal places:

a

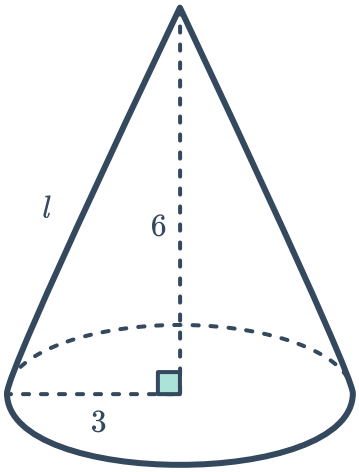


b



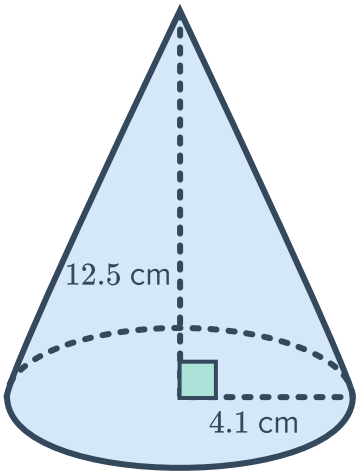
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c

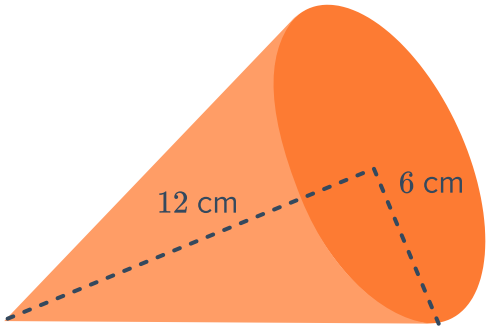


11 Find the volume of the following cones, correct to one decimal place:

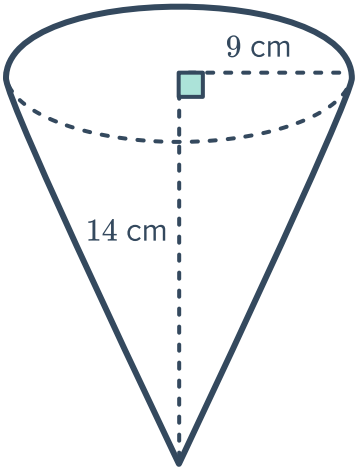
a



b



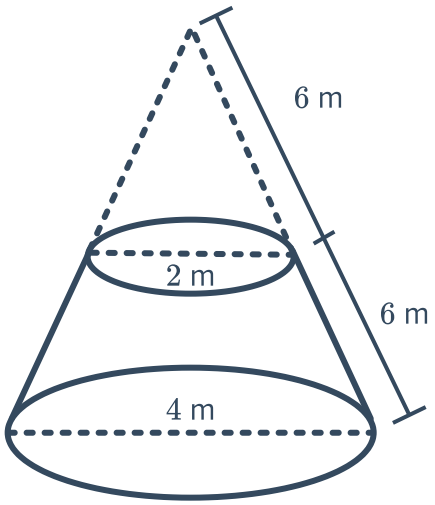
c



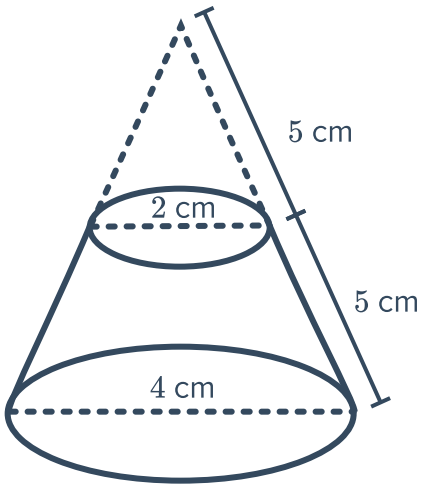
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12 The podium was formed by sawing off the top of a cone. Find the volume of the following podiums, correct to two decimal places:

a



b



13 Find the volume of the following cones, correct to two decimal places:

a A cone with radius of $8\ \text{cm}$ and vertical height of $13\ \text{cm}$.

b A cone with radius of $60\ \text{m}$ and a vertical height of $27.54\ \text{m}$.

14 Find the height of the following cones:

a A cone with volume $603.19\ \text{m}^3$ and radius $6\ \text{m}$. Round your answer to one decimal place.

b A cone with volume $26\,640.97\ \text{cm}^3$ and radius $31.9\ \text{cm}$. Round your answer to the nearest cm.

15 Find the radius of the following cones:

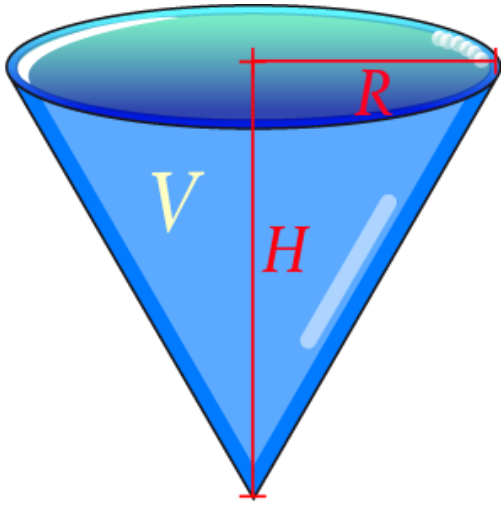
a A cone with volume of $12\,441.02\ \text{cm}^3$ and a height of $30\ \text{cm}$. Round your answer to one decimal place.

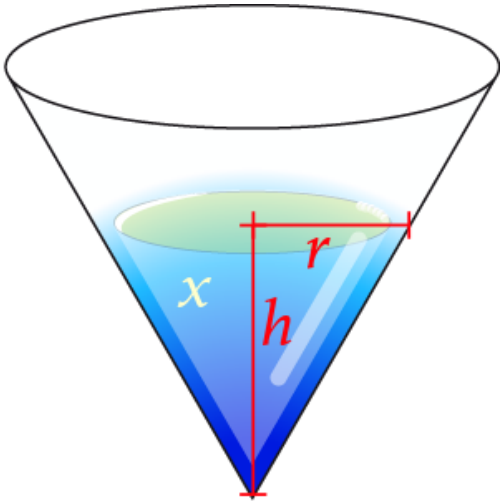
b A cone that has volume $1273.39\ \text{mm}^3$ and height $19\ \text{mm}$. Round your answer to the nearest mm.

c A cone with volume of $196\ \text{mm}^3$ and the length of the height and radius are equal. Round your answer to two decimal places.

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16 A water vessel is produced in the shape of a cone of radius R and maximum volume V.
From vertex to base, the vessel has a height of H.
A volume x of water is poured into the vessel, filling up from the vertex towards the base.
The radius of the cone of water within the vessel has radius r and height h.



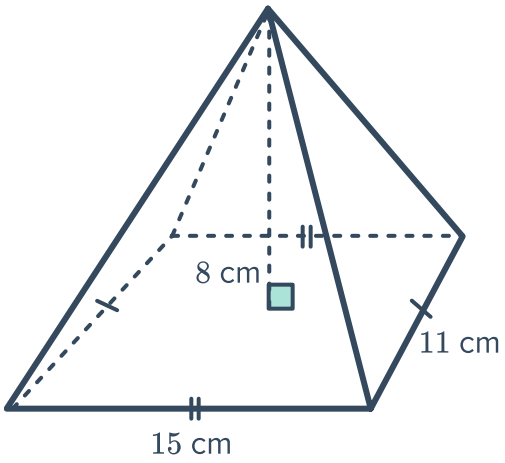


Write an expression for the height of the water, h, in terms of the height of the vessel H and the volumes V and x.

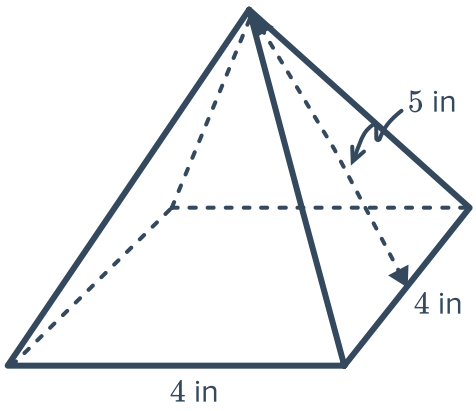
Additional questions

17 Find the volume of the following solids, rounding to two decimal places if necessary:

a

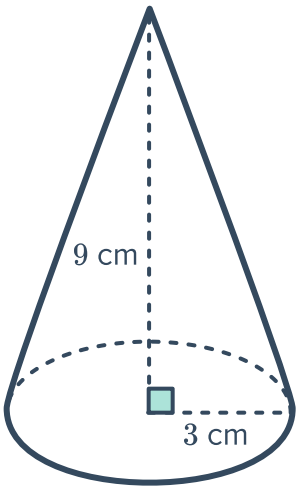


b

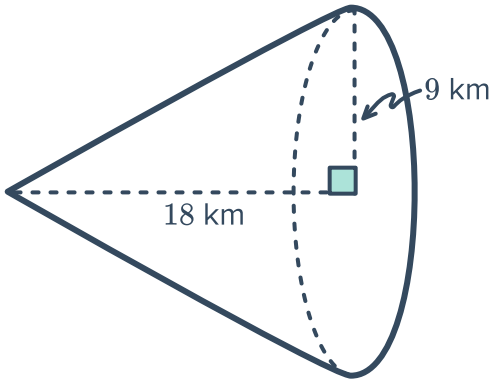


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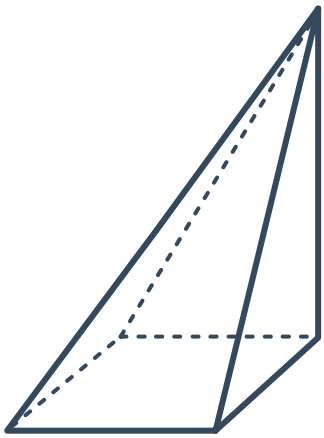
c



d



18 The volume of a right square pyramid is $55\ \text{in}^2$. If the pyramid is slanted by 60°, find its volume.

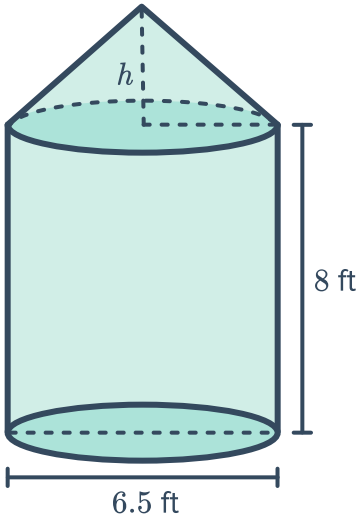


19 The radius and the height of a cone are equal, and the volume of the cone is 72π. Find the radius of the cone.

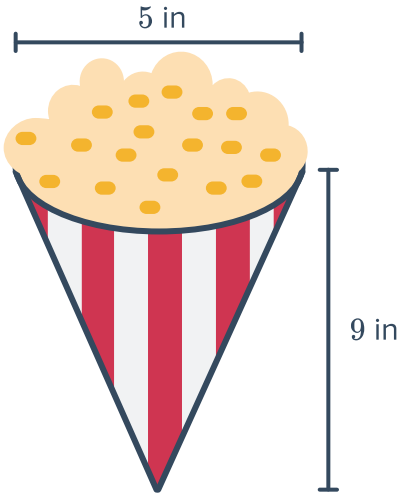
20 The volume of a pyramid is $60\ \text{m}^3$ with a height of $4\ \text{m}$. Find the area of the base.

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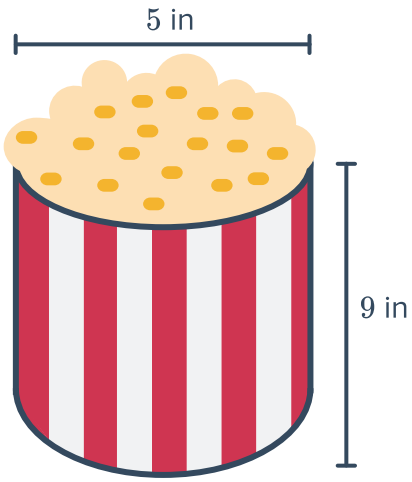
21 The Florida Agricultural Museum plans to bring in a replica grain silo for a new exhibit. The volume of the conical portion of the silo is $\frac{169}{24}\pi$ ft^3.
The museum's doorway has a height clearance of 12 ft . Will the model grain silo fit through the door? Explain.



22 A vendor at the Sumter County Fair sells kettlecorn in two containers.
Option 1 sells for $\$3.50$:



Option 2 sells for $\$10.50$:



If you have $\$12.00$ to spend, what would you order to get the maximum amount of kettlecorn? Explain.

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